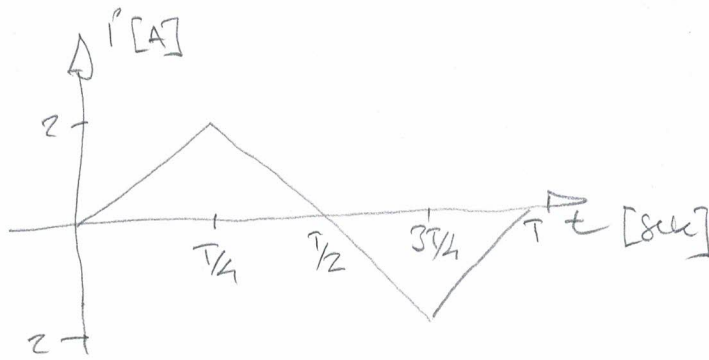


4.33



vi vet att $i_{rms} = \sqrt{\frac{1}{T} \int_0^T i(t)^2 dt}$

men kurvan kan vi dela upp i fyra lika stora delar

använder första delen

$$i(t) = \frac{8}{T}t, \quad 0 \leq t \leq \frac{T}{4}$$

$$\begin{aligned} \text{vi får} \quad i_{rms}^2 &= \frac{1}{T} \int_0^{T/4} \left(\frac{8}{T}t\right)^2 dt = \frac{1}{T} \frac{64}{T^2} \left[\frac{1}{3}t^3\right]_0^{T/4} = \\ &= \frac{1}{T} \frac{64}{T^2} \frac{1}{3} \frac{T^3}{64} = \frac{1}{3} \end{aligned}$$

$$\Rightarrow i_{rms} = \frac{2}{\sqrt{3}} \text{ A}$$

Svar: $i_{rms} = \frac{2}{\sqrt{3}} \text{ A}$